

Constraint-Conditioned Heterogeneous Graph Policies for Safe IoT-Aware Truck Backhaul Recommendation

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Abstract—Empty truck returns are not only a shortest-path problem: the dispatcher must jointly decide which heterogeneous truck should serve which open load, whether either should wait, and whether uncertain telemetry makes a recommendation unsafe. We present BACKHAULGRAPHPOLICY, a 4.39-million-parameter heterogeneous graph policy trained from random initialization to score all truck–order edges jointly. It consumes 32-dimensional IoT/fleet truck features, 24-dimensional order features, and 16-dimensional pairwise road/economic features. Four relation-attention blocks produce assignment, contribution, uncertainty, and ETA estimates plus a learned WAIT action. Deterministic masks retain operational authority over capacity, vehicle compatibility, manifest binding, fuel, telemetry freshness, time windows, and road availability; a conflict-free decoder prevents duplicate assignments and all outputs require dispatcher acceptance. The frozen TorchScript model was trained for 6,000 steps in 360.4 s on one NVIDIA A100-SXM4-80GB. On 3,520 fixed-seed snapshots from an Indonesia-style synthetic digital twin, it improved mean simulated reward by 0.1107 (6.64%) under standard conditions and 0.0617 (9.13%) under sensor-dropout stress relative to a static margin–distance–time score, with zero decoded hard-constraint violations and zero duplicate assignments. These results demonstrate executable model behavior only; they are not evidence of real-world Haulio uplift. Production claims require joined historical outcomes, chronological validation, shadow deployment, and a controlled dispatcher trial.

Index Terms—backhaul optimization, graph neural network, Internet of Things, vehicle routing, pickup and delivery, safe AI, logistics

I. INTRODUCTION

The Haulio challenge targets a recurring heavy-truck inefficiency: a vehicle completes a delivery but returns without a paid load, while discovery of a compatible backhaul remains reactive and fragmented across marketplaces and manual brokerage [1]. Global Navigation Satellite System (GNSS) position alone does not solve the decision. A useful recommender must combine truck readiness and health, load and body compatibility, pickup and delivery windows, route conditions, future demand, margin, and the option to wait for a better order.

A small modification to A* is insufficient for this decision boundary. A* finds a least-cost path between fixed endpoints on a graph. Backhaul recommendation first selects endpoints and resources under one-to-one assignment, heterogeneity, uncertainty, and competing orders; a road router can then

supply distances and times for each candidate edge. Conversely, replacing every rule with a neural model is unsafe: legality, capacity, identity, and temporal feasibility should remain explicit, auditable constraints. We therefore use a learned policy for the part where data-driven generalization matters—the joint value of feasible truck–order edges—inside a deterministic safety envelope.

This paper makes four contributions:

- an end-to-end, randomly initialized heterogeneous graph policy that jointly scores a market snapshot rather than applying a rule list to each pair independently;
- an IoT-grounded feature contract in which every deployed input maps to truck telemetry, fleet master data, order/manifest data, or obtainable road and weather context;
- an immutable preliminary-round inference architecture with checksum verification, explicit abstention, dispatcher authority, and no training or automatic feedback path; and
- reproducible A100 training and fixed-seed synthetic evidence, reported with an explicit prohibition on interpreting simulator reward as field uplift.

The current system is deliberately narrower than a universal vehicle-routing solver. It recommends conflict-free full-truckload (FTL) backhaul assignments for at most 16 trucks and 32 open orders in one overlapping corridor snapshot. Multi-stop sequencing, regulatory route legality, and field-calibrated outcome prediction remain future work.

II. RELATED WORK AND THE OPERATIONAL GAP

The vehicle-routing problem (VRP) originated in dispatching gasoline trucks [2] and now covers capacities, time windows, pickups and deliveries, stochastic arrivals, and heterogeneous fleets [3]. Classical exact and heuristic methods remain indispensable, particularly for hard guarantees. Learned combinatorial optimization instead exploits repeated instances from a distribution. Pointer Networks [4], graph-embedding policies [5], and attention models [6] showed that neural policies can construct or improve routing solutions. Kool *et al.* train an attention model with REINFORCE and a rollout baseline across several stylized routing tasks [6]; Wu *et al.* learn an iterative improvement heuristic for TSP and CVRP [7].

Recent logistics work moves closer to dynamic operations. Li *et al.* combine spatial-temporal demand prediction with a

graph value network for an industry-scale dynamic pickup-and-delivery problem [8]. CAADRL explicitly encodes spatial clusters and uses hierarchical decoding for paired pickup/delivery nodes [9]. Its cluster-density inductive bias is promising when stops form local groups, but the reported task is a synthetic node-sequencing PDP. Heavy-truck backhaul additionally requires vehicle body compatibility, tonnage, cargo class, manifest identity, current commitment, device health, and uncertain availability. Those constraints cannot safely be left to implicit attention.

Our model consequently is neither “A* with a neural layer” nor an online reinforcement-learning agent. It is an offline policy learner: solver-style conflict-free targets, heteroscedastic outcome regression, ETA regression, and a differentiable expected-reward objective train one graph network from random initialization. At runtime only immutable inference is present. This hybrid separation follows a practical principle: learn the context-dependent ranking that hand-tuned coefficients miss, while compile-and-verify constraints remain deterministic.

III. DECISION PROBLEM

At decision epoch t , let $\mathcal{T}$ be active trucks and $\mathcal{O}$ open orders. Truck i has state x_i^T , order j has state x_j^O , and pair (i, j) has context x_{ij}^E . The runtime builds a bipartite graph with truck and order node types and candidate edges. A binary mask m_{ij} states whether an edge is feasible before model inference.

Let $z_{ij} = 1$ assign truck i to order j , while $z_{i0} = 1$ chooses WAIT. A valid recommendation satisfies

$$\sum_{j \in \mathcal{O}} z_{ij} + z_{i0} = 1, \quad \forall i \in \mathcal{T}, \quad (1)$$

$$\sum_{i \in \mathcal{T}} z_{ij} \leq 1, \quad \forall j \in \mathcal{O}, \quad (2)$$

$$z_{ij} \leq m_{ij}. \quad (3)$$

The learned policy produces edge logit ℓ_{ij} , log variance s_{ij} , utility $\hat{u}_{ij}$, ETA $\hat{t}_{ij}$, and truck-level wait logit ℓ_{i0} . The deployed edge score is

$$q_{ij} = \ell_{ij} - 0.10 \exp(s_{ij}/2), \quad (4)$$

so uncertain edges are conservatively discounted. An edge is removed if predicted ETA exceeds its delivery service-level agreement (SLA). The decoder sorts positive edge advantages $q_{ij} - \ell_{i0}$ and greedily accepts an edge only if neither endpoint has been used. This decoder is a fast matching heuristic, not a claim of global optimality.

IV. DATA CONTRACT AND DIGITAL TWIN

A. IoT-Relevant Runtime Features

Table I maps the state to obtainable sources. Optional sensors never silently become zeros: CAN, IMU, cargo-weight, and gateway-health presence flags let the model distinguish a true zero from missing data. GNSS age, accuracy, and replay status preserve store-and-forward semantics. The identity and manifest join occurs outside the model so a plausible but unauthenticated packet cannot create a candidate.

The same contract can later ingest licensed marketplace or carrier histories, but such records are not included in this run. Public road geometry (e.g., OpenStreetMap [10]), public weather, and appropriately licensed mobility traces can support domain-shift pretraining or stress curricula; they cannot substitute for Haulio’s joined *decision–acceptance–realized outcome* labels. The current evidence therefore uses only labelled synthetic data.

B. Indonesia-Style Synthetic Curriculum

The vectorized generator samples eight spatial hubs, 4–16 trucks, and 8–32 orders per snapshot. It includes five truck/body and five cargo categories, capacities between roughly 8 and 18 tonnes before perturbation, pickup and delivery windows, cancellation probability, price, toll, weather, congestion, road quality, device replay, and unavailable roads. Standard generation drops each optional sensor family with probability 0.12; the stress slice raises dropout to 0.34, extends telemetry ages, increases replay frequency, and increases missing manifests and unavailable roads.

For feasible pair (i, j) the simulated reward is

$$r_{ij} = \tilde{M}_{ij} + 0.42D_j + 0.30A_{ij} - 1.10L_{ij} \quad (5)$$

$$- 0.55H_i - 0.24U_i - 0.34C_j + 0.10P_j + \epsilon, \quad (6)$$

where $\tilde{M}$ is normalized margin, D destination demand density, A home alignment, L lateness, H health risk, U telemetry uncertainty, C cancellation probability, P priority, and ϵ simulated noise. A separate wait reward depends on local future demand and health. A conflict-free greedy oracle constructs training actions from these simulated rewards. This curriculum exposes interactions and failures efficiently, but it also embeds designer assumptions. It must not be described as observed customer behavior.

V. MODEL AND SYSTEM ARCHITECTURE

Fig. 1 shows the preliminary runtime. IoT data travel through the automotive store-and-forward gateway and the existing MQTT/NestJS ingestion path. Server-side validation updates operational state. A bounded corridor snapshot and road/weather context feed both a deterministic constraint compiler and the frozen policy. There is no arrow from dispatcher decisions back to weights.

A. Heterogeneous Relation Network

The network in Fig. 2 embeds $x_i^T \in \mathbb{R}^{32}$ and $x_j^O \in \mathbb{R}^{24}$ independently into width $d = 192$. In each of four relation layers, a pair MLP embeds $x_{ij}^E \in \mathbb{R}^{16}$ as r_{ij} . A learned gate scores

$$a_{ij} = w^\top \tanh(h_i^T + h_j^O + r_{ij}). \quad (7)$$

Masked softmax over orders aggregates order-plus-relation messages into every truck; softmax over trucks symmetrically updates every order. Residual MLPs and LayerNorm produce the next states. This bidirectional message passing lets one edge’s score depend on competing trucks and orders, which an independent pair rule cannot express.

TABLE I
RUNTIME DATA CONTRACT. “FALLBACK” DESCRIBES INFERENCE-TIME BEHAVIOR, NOT MODEL UPDATING.

Source	Representative fields	Operational relevance	Missing/stale fallback
GNSS and gateway	latitude, longitude, speed, heading, accuracy, event time, replay flag, signal, power, uptime	position, readiness, deadhead, direction, confidence	telemetry older than 300 s is infeasible; accuracy and replay reduce quality
CAN/J1939 (optional)	fuel, engine speed, coolant and presence flag	fuel gate and vehicle-health risk	missing CAN is encoded; it is not fabricated
IMU (optional)	acceleration, gyroscope and presence flag	abnormal-motion and quality context	missing IMU is explicitly encoded
Cargo interface	empty/load ratio, weight observation and presence flag	true availability and capacity state	unknown reading lowers confidence; manifest/capacity gates still apply
Fleet/TMS	truck ID, body type, rated capacity, home, commitment/available-after	heterogeneous compatibility and home-bound value	unknown vehicle type is rejected
Order/manifest	pickup/drop-off, weight, cargo class, required vehicle, windows, offer, priority, service time, manifest binding	feasibility, economic value and SLA	unknown class, unbound manifest, or missed window is rejected
Road/weather context	deadhead/loaded km, travel time, toll, road quality, congestion, weather, route freshness	ETA, operating cost and disruption risk	unavailable road is rejected; stale context is flagged
Demand context	density of open pickups near destination	chance of useful downstream continuation	computed from the current order book; no external personal data

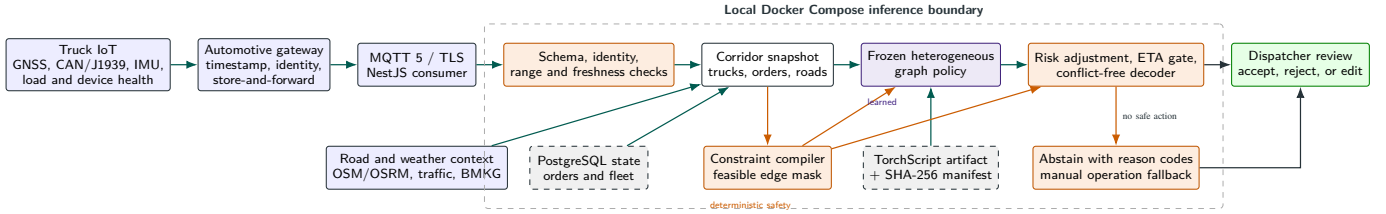


Fig. 1. End-to-end inference architecture. Violet denotes the learned component; orange denotes deterministic safety. The preliminary submission contains static core inference only, and the dispatcher remains the final authority.

After four layers, the model concatenates $[h_i^T, h_j^O, r_{ij}] \in \mathbb{R}^{576}$ and applies separate policy, utility, log-variance, and positive-ETA heads. A truck-state head produces WAIT . The complete model has 4,393,161 trainable parameters.

B. Safety Envelope and Edge Cases

The system treats abstention as a valid output. Table II lists implemented runtime behavior. A complete production plan would additionally require route-leg legality, driver hours, axle constraints, temperature-chain state, ferry schedules, toll/payment limitations, facility calendars, and emergency overrides; those fields are not claimed by this checkpoint.

VI. TRAINING AND FREEZING

Training started from random initialization; no downloaded checkpoint or pretrained encoder was used. The loss at step k was

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + 0.34\mathcal{L}_{\text{NLL}} + 0.16\mathcal{L}_{\text{ETA}} \quad (8)$$

$$+ \omega_k (-\mathbb{E}[r] + 0.45\mathcal{L}_{\text{collision}} - 0.006\mathcal{H}), \quad (9)$$

where $\mathcal{L}_{\text{CE}}$ imitates the conflict-free oracle including WAIT , $\mathcal{L}_{\text{ETA}}$ is Smooth- L_1 on $\log(1 + \text{ETA})$, and the heteroscedastic utility loss is

$$\mathcal{L}_{\text{NLL}} = \frac{1}{2} \text{mean} [e^{-s_{ij}} (\hat{u}_{ij} - r_{ij})^2 + s_{ij}]. \quad (10)$$

TABLE II
IMPLEMENTED DETERMINISTIC SAFETY BEHAVIOR.

Condition	Runtime action
Unknown truck/cargo type	reject pair with reason code
Weight exceeds capacity available at pickup	reject pair
Fuel below 10%	reject pair
Telemetry age above 300 s	reject pair
Manifest not bound	reject pair
Pickup/delivery window missed	reject pair
Road unavailable	reject pair
Predicted ETA above SLA	reject pair after inference
Truck/order already selected	skip edge in decoder
No positive safe edge	abstain; manual fallback
Artifact/manifest mismatch	fail startup checksum gate
Any recommendation	require human acceptance; never auto-commit

The direct policy term begins after warmup and ramps to $\omega_k = 0.30$. It maximizes expected simulated reward, discourages multiple trucks placing mass on one order, and retains small entropy. This is differentiable policy optimization, not a deployed online-RL loop.

AdamW used learning rate 3×10^{-4} , weight decay 10^{-2} , $\beta = (0.9, 0.95)$, batch size 256, 400 warmup steps followed by cosine decay, BF16 autocast, and gradient-norm clipping

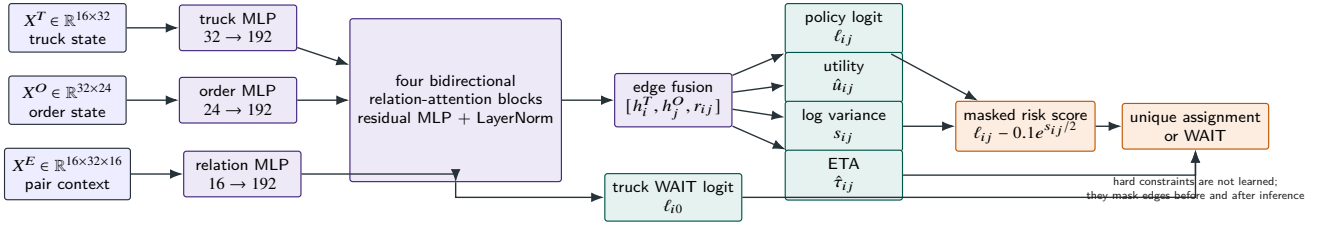


Fig. 2. Frozen learned core and output heads. Node/pair padding masks support bounded variable-size snapshots. Feasibility is compiled independently and applied to policy logits and final decoded assignments.

TABLE III
FIXED-SEED SYNTHETIC RESULTS. REWARD IS SIMULATOR-DEFINED AND UNITLESS.

Metric	Standard	Dropout stress
Static baseline reward	1.6661	0.6764
Learned reward	1.7768	0.7381
Absolute difference	+0.1107	+0.0617
Relative difference	+6.64%	+9.13%
Teacher reward	1.8178	0.7535
Teacher action agreement	96.20%	98.17%
A100 model-only p95 (ms/snapshot)	0.1309	0.1111
Hard violations	0	0
Duplicate assignments	0	0

at 1.0. Sensor-dropout stress probability increased to 0.35 over training. A combined standard-plus-stress metric selected the checkpoint at step 4,000; optimization completed 6,000 steps in 360.4117 s on an NVIDIA A100-SXM4-80GB with PyTorch 1.13.1/CUDA 11.6. The invoked time budget was four hours, with independent script and process limits below ten hours.

The best model was copied to CPU FP32, traced, frozen, and saved as TorchScript. The recorded eight-snapshot export check had maximum absolute difference 0.0; an independent 30-snapshot audit measured 7.63×10^{-6} , below the 10^{-5} parity gate. The 17,609,009-byte artifact is bound to manifest version 2026.08.25-a100. Its SHA-256 digest is

c33c4a83ac3a6925 cc9634ecf6d29fb9
29427918ed70440c f8560beb3527b733

VII. SYNTHETIC EVALUATION

The final fixed-seed evaluation uses 20 standard batches of 128 snapshots and 10 stress batches of 96, totaling 3,520. The static baseline greedily decodes $2.1\tilde{M} - 0.35\tilde{d}_{\text{deadhead}} - 0.18\tilde{t}_{\text{road}}$ under the same feasibility mask and uniqueness rule. The teacher is the conflict-free oracle over simulated realized reward.

Table III shows higher mean simulated reward than the static score in both slices, while retaining zero decoded hard violations and duplicates. Stress reward is lower for every policy because the generator deliberately creates more missing, stale, and infeasible observations. The unexpectedly lower model-only latency in the stress slice is within benchmark variation and should not be interpreted as a robustness benefit.

The experiment does *not* establish statistical significance, optimality, cross-seed stability, domain generalization, safety

in traffic, or financial return. The oracle and reward arise from the same simulator, so close teacher agreement may partly reflect learning simulator structure. No hidden or proprietary Haulio labels were used.

VIII. PRELIMINARY DEPLOYMENT AND COMPLIANCE

The competition bundle cleanly separates development from demonstration. The submission directory contains the frozen artifact, static manifest and metrics, deterministic feature builder/decoder, a minimal HTTP service, and Docker Compose. It exposes only /health and /infer on loop-back port 8088. The container uses a read-only root filesystem, a bounded temporary filesystem, a non-root user, and no-new-privileges. The inference response records the model hash, reason-coded rejected pairs, uncertainty, abstention, automatic_commit=false, and requires_human_acceptance=true.

Changing telemetry and orders changes inference inputs but never parameters. The runtime contains no optimizer, backward pass, automatic tuning, bulk testing, feedback loop, model promotion, or network data collection. This satisfies the preliminary requirement that only core inference with static parameters be demonstrated. The explanatory notebook is not a runtime dependency.

Reproduction is supported by the offline command ./run_a100.sh, fixed seed 20260825, recorded environment and training history, the unexecuted and executed demonstration notebooks, manifest, metrics, and checksum. Reproducing the exact GPU timing still depends on compatible A100/CUDA/PyTorch hardware and software.

IX. LIMITATIONS AND FIELD-VALIDATION PLAN

The central limitation is data provenance: the model has not seen joined Haulio order, recommendation, dispatcher decision, route, and realized outcome histories. Synthetic training makes the system executable under a ten-hour budget, but it cannot validate the business objective. A production program should: (1) define event-time joins and reason-coded human decisions; (2) replay historical snapshots chronologically without future leakage; (3) calibrate ETA and uncertainty by corridor, truck type, and missingness pattern; (4) compare against strong OR/heuristic baselines and solver upper bounds; (5) run a no-commit shadow deployment; and (6) conduct a controlled dispatcher trial with safety, service, empty-kilometre, contribution-margin, and acceptance metrics.

The present bounded graph also omits multi-hop route construction and paired pickup/delivery precedence. Markets larger than 16×32 require overlapping corridor partitioning, which can miss cross-partition assignments. There is no guarantee under severe distribution shift, and the reported OOD score is diagnostic rather than a calibrated rejection probability. Finally, some relevant heavy-vehicle constraints—axle loads, driver hours, permits, hazmat route restrictions, cold-chain temperature, facility calendars, and road-class legality—require new authoritative fields and validators before field use.

X. CONCLUSION

We implemented an actual learned core rather than relabelling A* or a rule table as AI. The heterogeneous graph policy learns contextual competition among trucks and orders, contribution, uncertainty, ETA, and a wait decision from random initialization; deterministic code retains non-negotiable operational constraints. The frozen model executes quickly and exceeds a static scoring baseline in a labelled synthetic digital twin, including sensor-dropout stress, with no decoded violations in the evaluated snapshots. The honest conclusion is correspondingly narrow: the architecture and artifact are ready for a competition demonstration and for shadow-data

collection, not for claims of autonomous or real-world optimization. Human approval, immutable preliminary inference, and rigorous field validation remain essential.

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